1 Space Curves and Frenet Frames

1.1 Curvature and Frenet Frame

• Let $\mathbf{r}(s)$ be a space curve in $\mathbb{R}^3$, where s is the parameter of arc-length, hence we have $|\mathbf{r}'(s)| \equiv 1$.

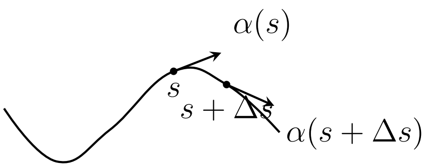
Notation: Let $\mathbf{r}(s)$ be a curve in $\mathbb{R}^3$ with s being the parameter of arc-length, we denote

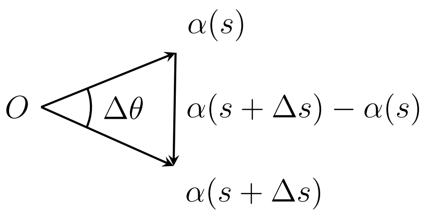
$$\alpha(s) := \mathbf{r}'(s) \triangleq \frac{\mathrm{d}}{\mathrm{d}s}\mathbf{r}(s),$$

which also implies that $|\alpha(s)| \equiv 1$.

Proposition 1. Let $\Delta\theta := \langle \alpha(s+\Delta s), \alpha(s) \rangle$, then we have

$$\left|\frac{\mathrm{d}\alpha}{\mathrm{d}s}\right| = \lim_{\Delta s \to 0} \left|\frac{\Delta\theta}{\Delta s}\right|.$$





Proof: We only need to note that

$$\left|\frac{\mathrm{d}\alpha}{\mathrm{d}s}\right| = \lim_{\Delta s \to 0} \left|\frac{\alpha(s+\Delta s)-\alpha(s)}{\Delta s}\right| = \lim_{\Delta s \to 0} \left|\frac{2\sin\left(\frac{\Delta\theta}{2}\right)}{\Delta s}\right| = \lim_{\Delta s \to 0} \frac{2\left|\left(\frac{\Delta\theta}{2}\right)\right|}{|\Delta s|} = \lim_{\Delta s \to 0} \left|\frac{\Delta\theta}{\Delta s}\right|,$$

where we used the fact that $\sin t \sim t$ as $t \to 0$. ■

• Recall that for a curve $\mathbf{r}(s)$ with arc-length parameter, we have defined its curvature at the point $\mathbf{r}(s)$ to be

$$\kappa(s) := |\mathbf{r}''(s)| = \left|\frac{\mathrm{d}^2\mathbf{r}(s)}{\mathrm{d}s^2}\right| = |\alpha'(s)|,$$

and $\alpha'(s) = \frac{\mathrm{d}\alpha}{\mathrm{d}s}$ is called the ***curvature vector*** of the curve.

• Recall that for a regular curve $\mathbf{r}(t)$, its arc-length element $\mathrm{d}s$ is given by $\mathrm{d}s = |\mathbf{r}'(t)|\mathrm{d}t$.

Definition 1. The ***tangent image*** (or ***tangent indicatrix***) of a curve $C : \mathbf{r} = \mathbf{r}(s)$ is the curve traced out by the endpoint of its unit tangent vector $\alpha(s) = \mathbf{r}'(s)$ when it is translated parallelly to the origin point O.

Remark: Since the parametric equation for the tangent image of the curve $\mathbf{r}(s)$ is $\widetilde{\mathbf{r}} = \alpha(s)$, so its arc-length element $\mathrm{d}\widetilde{s}$ is given by

$$\mathrm{d}\widetilde{s} = \left|\frac{\mathrm{d}\alpha(s)}{\mathrm{d}s}\right| \mathrm{d}s = \kappa(s)\mathrm{d}s,$$

so we see that

$$\kappa(s) = \frac{\mathrm{d}\widetilde{s}}{\mathrm{d}s},$$

which means that the curvature $\kappa(s)$ of the curve is the ratio of these two arc-length elements.

• For the curve $C : \mathbf{r} = \mathbf{r}(s)$, since $|\alpha(s)| = |\mathbf{r}'(s)| \equiv 1$, so we get $\frac{\mathrm{d}}{\mathrm{d}s}(\alpha(s)\cdot\alpha(s)) = 2\alpha(s)\cdot\alpha'(s) \equiv 0$, i.e., $\alpha(s)\cdot\alpha'(s) \equiv 0$ for $\forall s$, hence $\alpha(s) \perp \alpha'(s)$, which means the vector $\alpha'(s)$ is a normal vector of the curve $C : \mathbf{r} = \mathbf{r}(s)$.

Definition 2.(Princial Normal Vector)
For a curve $C : \mathbf{r} = \mathbf{r}(s)$, where s is the arc-length parameter, if thec curvature $\kappa(s) := |\alpha'(s)| \neq 0$, then the vector $\beta(s)$, which is defined to be

$$\beta(s) := \frac{\alpha'(s)}{\kappa(s)} = \frac{\mathbf{r}''(s)}{|\mathbf{r}''(s)|}$$

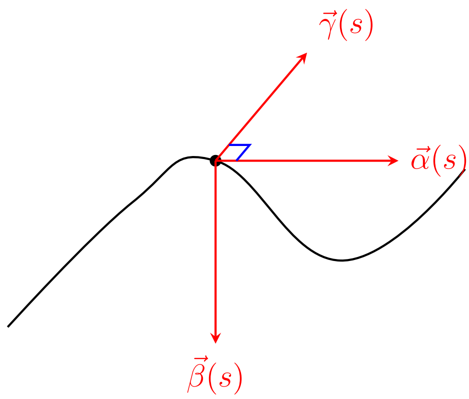
is said to be the ***principal normal vector*** of the curve $C : \mathbf{r} = \mathbf{r}(s)$.

Remark: According to the definition above, we see that the princal normal vector $\beta(s)$ can be written as

$$\alpha'(s) = \kappa(s)\beta(s).$$

Definition 3.(Binormal Vector) The binormal vector $\gamma(s)$ of the space curve $C : \mathbf{r} = \mathbf{r}(s)$ is defined to be

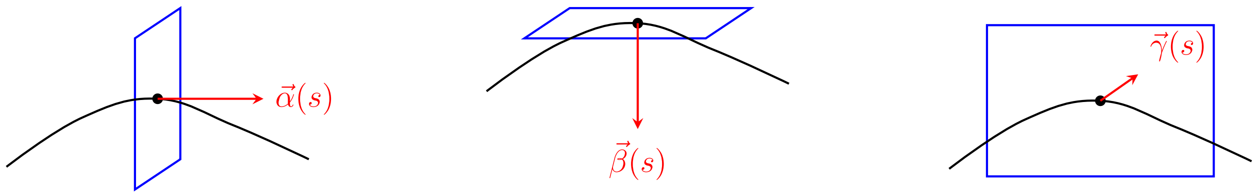
$$\gamma(s) := \alpha(s) \times \beta(s).$$



Remark: By these definitions above, we can obtain the ***Frenet frame*** $\{\mathbf{r}(s); \alpha(s), \beta(s), \gamma(s)\}$, which is a right-handed orthonormal frame. If $\kappa(s_0) = 0$, then the Frenet frame $\{\mathbf{r}(s_0); \alpha(s_0), \beta(s_0), \gamma(s_0)\}$ at the point $\mathbf{r}(s_0)$ is not defined.

Definition 4. For the curve $C : \mathbf{r} = \mathbf{r}(s)$, if the curvature $\kappa(s) \neq 0$, then the three lines through a point of the curve, directed along the tangent vector $\alpha(s)$, the principal normal vector $\beta(s)$, and the binormal vector $\gamma(s)$, are respectively called the ***tangent line***, the ***principal normal line***, and the ***binormal line*** of the curve at that point.

Definition 5.
(1) The plane passing through the point $\mathbf{r}(s)$ and having $\alpha(s)$ as its normal vector is called the ***normal plane***;
(2) The plane passing through the point $\mathbf{r}(s)$ and having $\beta(s)$ as its normal vector is called the ***rectifying plane***;
(3) The plane passing through the point $\mathbf{r}(s)$ and having $\gamma(s)$ as its normal vector is called the ***osculating plane***.



Remark: By the definition, we can easily gain the equation for these three planes
(a) **normal plane**: $(\mathbf{X} - \mathbf{r}(s)) \cdot \alpha(s) = 0$.
(b) **rectifying plane**: $(\mathbf{X} - \mathbf{r}(s)) \cdot \beta(s) = 0$.
(c) **osculating plane**: $(\mathbf{X} - \mathbf{r}(s)) \cdot \gamma(s) = 0$,
where $\mathbf{X}$ is the position vector of the poits on the plane.

We give the formula of the curvature and Frenet frame.

Proposition 2.(Formula of the Curvature and Frenet Frame)
(1) If $\mathbf{r}(s)$ is a space curve with arc-length parameter, then we have

$$\kappa(s) = \left| \frac{\mathrm{d}^2 \mathbf{r}(s)}{\mathrm{d}s^2} \right| = |\alpha'(s)| \text{ and } \begin{cases} \alpha(s) = \mathbf{r}'(s); \\ \beta(s) = \frac{\mathbf{r}''(s)}{|\mathbf{r}''(s)|}; \\ \gamma(s) = \alpha(s) \times \beta(s) = \frac{\mathbf{r}'(s) \times \mathbf{r}''(s)}{|\mathbf{r}''(s)|}. \end{cases}$$

(2) If $\mathbf{r}(t)$ is a regular curve where t might not be the arc-length parameter, then we have

$$\kappa(t) = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3},$$

and

$$\begin{cases} \alpha(t) = \frac{\mathbf{r}(t)}{|\mathbf{r}(t)|}; \\ \gamma(t) = \frac{\mathbf{r}'(t) \times \mathbf{r}''(t)}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}; \\ \beta(t) = \gamma(t) \times \alpha(t) = \frac{|\mathbf{r}'(t)|}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|} \mathbf{r}''(t) - \frac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{|\mathbf{r}'(t)||\mathbf{r}'(t) \times \mathbf{r}''(t)|} \mathbf{r}'(t). \end{cases}$$

Proof: (1) By definition.

(2) It's clear that $\alpha(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$, hence $\mathbf{r}'(t) = |\mathbf{r}'(t)|\alpha(t)$, thus

$$\begin{aligned}
\mathbf{r}''(t) &= \frac{\mathrm{d}}{\mathrm{d}t}\mathbf{r}'(t) \\
&= \frac{\mathrm{d}}{\mathrm{d}t}\Big(|\mathbf{r}'(t)|\alpha(t)\Big) = \frac{\mathrm{d}|\mathbf{r}'(t)|}{\mathrm{d}t}\alpha(t) + |\mathbf{r}'(t)|\frac{\mathrm{d}}{\mathrm{d}t}\alpha(t) \\
&= \frac{\mathrm{d}|\mathbf{r}'(t)|}{\mathrm{d}t}\alpha(t) + |\mathbf{r}'(t)|\frac{\mathrm{d}\alpha(t)}{\mathrm{d}s}\frac{\mathrm{d}s}{\mathrm{d}t} \\
&= \frac{\mathrm{d}|\mathbf{r}'(t)|}{\mathrm{d}t}\alpha(t) + |\mathbf{r}'(t)|^2 \cdot \kappa(t)\beta(t) \quad \Big(\text{since } \alpha'(s) = \kappa(s)\beta(s) \text{ and } s'(t) = |\mathbf{r}'(t)|\Big).
\end{aligned}$$

Thus we get

$$\begin{aligned}
\mathbf{r}'(t) \times \mathbf{r}''(t) &= \Big(|\mathbf{r}'(t)|\alpha(t)\Big) \times \Big(\frac{\mathrm{d}|\mathbf{r}'(t)|}{\mathrm{d}t}\alpha(t) + |\mathbf{r}'(t)|^2 \cdot \kappa(t)\beta(t)\Big) \\
&= |\mathbf{r}'(t)|^3 \cdot \kappa(t)\Big(\alpha(t) \times \beta(t)\Big) \\
&= |\mathbf{r}'(t)|^3 \cdot \kappa(t)\gamma(t) \quad (\text{since } \gamma(t) = \alpha(t) \times \beta(t) \text{ by definition}).
\end{aligned}$$

This means that

$$|\mathbf{r}'(t) \times \mathbf{r}''(t)| = |\mathbf{r}'(t)|^3\kappa(t),$$

i.e.,

$$\kappa(t) = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3}.$$

Lastly, the equation that

$$\beta(t) = \gamma(t) \times \alpha(t) = \frac{|\mathbf{r}'(t)|}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}\mathbf{r}''(t) - \frac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{|\mathbf{r}'(t)||\mathbf{r}'(t) \times \mathbf{r}''(t)|}\mathbf{r}'(t)$$

is gained by inserting the equations of $\gamma(t)$ and $\alpha(t)$ respectively. ■

Example 1. Consider the circular helix $\mathbf{r}(t) = (a\cos t, a\sin t, bt)$ with $a, b \in \mathbb{R}^+$ constants. Find the arc-length parametrization of $\mathbf{r}(t)$ and compute its Frenet frame?

Proof: By $\mathbf{r}(t) = (a\cos t, a\sin t, bt)$, we obtain

$$\mathbf{r}'(t) = (-a\sin t, a\cos t, b),$$

$$\mathbf{r}''(t) = (-a\cos t, -a\sin t, 0),$$

$$\begin{aligned}
\mathbf{r}'(t) \times \mathbf{r}''(t) &= \left(\det\begin{pmatrix} a\cos t & b \\ -a\sin t & 0\end{pmatrix}, -\det\begin{pmatrix} -a\sin t & b \\ -a\cos t & 0\end{pmatrix}, \det\begin{pmatrix} -a\sin t & a\cos t \\ -a\cos t & -a\sin t\end{pmatrix}\right) \\
&= \Big(ab\sin t, -ab\cos t, a^2\Big).
\end{aligned}$$

Hence

$$\kappa(t) = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3} = \frac{a}{a^2+b^2},$$

and

$$\alpha(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \left(\frac{-a}{\sqrt{a^2+b^2}}\sin t, \frac{a}{\sqrt{a^2+b^2}}\cos t, \frac{b}{\sqrt{a^2+b^2}}\right),$$

$$\gamma(t) = \frac{\mathbf{r}'(t) \times \mathbf{r}''(t)}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|} = \left(\frac{b}{\sqrt{a^2+b^2}} \sin t, \frac{-b}{\sqrt{a^2+b^2}} \cos t, \frac{a}{\sqrt{a^2+b^2}} \right),$$
$$\beta(t) = \gamma(t) \times \alpha(t) = (-\cos t, -\sin t, 0).$$

More over, we obtained

$$s'(t) = |\mathbf{r}'(t)| = \sqrt{a^2+b^2},$$

which implies that

$$s = s(t) = \int_0^t s'(\tau) \mathrm{d}\tau = \sqrt{a^2+b^2} \cdot t \text{ and } t = t(s) = \frac{s}{\sqrt{a^2+b^2}},$$

so the arc-length parametrization of $\mathbf{r}(t)$ is given by

$$\mathbf{r} = \mathbf{r}(s) = \left(a \cos \left(\frac{s}{\sqrt{a^2+b^2}} \right), a \sin \left(\frac{s}{\sqrt{a^2+b^2}} \right), \frac{b}{\sqrt{a^2+b^2}} s \right).$$

1.2 Torsion and Frenet Formula

Idea: Recall that we have defined the osculating plane for a space curve $C : \mathbf{r} = \mathbf{r}(s)$, which is having $\gamma(s)$ as its normal plane. So if the space curve $C : \mathbf{r} = \mathbf{r}(s)$ is a plane curve, then this plane is actually the osculating plane of the curve whose binormal vector $\gamma(s)$ should be a constant vector. So based on this idea, $|\gamma'(s)|$ can be used to measure the rate of rotation of the osculating plane of the curve, i.e., the extent to which the curve deviates from being a plane curve, i.e., the so-called ***torsion***.

- Let $\gamma(s)$ be the binormal vector for the space curve $C : \mathbf{r} = \mathbf{r}(s)$, since $|\gamma(s)| \equiv 1$, so $\gamma'(s) \perp \gamma(s)$ for $\forall s$. By defintion, $\gamma(s) = \alpha(s) \times \beta(s)$, thus

$$\begin{aligned} \gamma'(s) &= \frac{\mathrm{d}}{\mathrm{d}s} \gamma(s) = \frac{\mathrm{d}}{\mathrm{d}s} \Big(\alpha(s) \times \beta(s) \Big) \\ &= \alpha'(s) \times \beta(s) + \alpha(s) \times \beta'(s) \\ &= \alpha(s) \times \beta'(s) \quad (\text{since } \alpha'(s) \parallel \beta(s)), \end{aligned}$$

which implies that $\gamma'(s) \perp \alpha(s)$ and $\gamma'(s) \perp \gamma(s)$, so this tells us that $\gamma'(s) \parallel \beta(s)$, thus we can write that

$$\gamma'(s) \triangleq \Big(-\tau(s) \Big) \cdot \beta(s),$$

i.e.,

$$\tau(s) = -\gamma'(s) \cdot \beta(s) \text{ and } |\tau(s)| = |\gamma'(s)|.$$

Definition 6.(Torsion of a Curve)
Let $\beta(s)$ and $\gamma(s)$ be the principal normal vector and the binormal vector of a curve $C : \mathbf{r} = \mathbf{r}(s)$ with arc-length parametrization, the function

$$\tau(s) := -\gamma'(s) \cdot \beta(s)$$

is called the ***torsion*** of the curve at the point $\mathbf{r}(s)$.

Theorem 1.(Frenet Formula)
The formula of the Frenet frame $\{\mathbf{r}(s); \alpha(s), \beta(s), \gamma(s)\}$ can be given by

$$\begin{pmatrix} \alpha'(s) \\ \beta'(s) \\ \gamma'(s) \end{pmatrix} = \begin{pmatrix} 0 & \kappa(s) & 0 \\ -\kappa(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{pmatrix} \begin{pmatrix} \alpha(s) \\ \beta(s) \\ \gamma(s) \end{pmatrix}.$$

Proof: We only need to find the formula for $\beta'(s)$. Note that

$$\beta'(s) \in \mathrm{Span}_{\mathbb{R}}\{\alpha(s), \beta(s), \gamma(s)\},$$

so we may denote

$$\beta'(s) \triangleq a\alpha(s) + b\beta(s) + c\gamma(s) \text{ with } a, b, c \in \mathbb{R}.$$

So we get

$$\begin{aligned} a &= \beta'(s) \cdot \alpha(s) \\ &= -\beta(s) \cdot \alpha'(s) \quad \left(\text{since } \frac{\mathrm{d}}{\mathrm{d}s}\Big(\alpha(s) \cdot \beta(s)\Big) \equiv 0\right) \\ &= -\kappa(s) \quad \Big(\text{ since } \alpha'(s) = \kappa(s)\beta(s)\Big), \end{aligned}$$

$$b = \beta'(s) \cdot \beta(s) = 0 \quad \left(\text{since } \frac{\mathrm{d}}{\mathrm{d}s}\Big(\beta(s) \cdot \beta(s)\Big) \equiv 0\right),$$

$$\begin{aligned} c &= \beta'(s) \cdot \gamma(s) \\ &= -\beta(s) \cdot \gamma'(s) \quad \left(\text{since } \frac{\mathrm{d}}{\mathrm{d}s}\Big(\beta(s) \cdot \gamma(s)\Big) \equiv 0\right) \\ &= \tau(s) \quad \Big(\text{since } \gamma'(s) = -\tau(s)\beta(s)\Big). \end{aligned}$$

So we see that

$$\beta'(s) = -\kappa(s)\alpha(s) + \tau(s)\gamma(s),$$

which completes the proof. ■

Remark: In general, for any orthonormal frame field along a curve, the matrix of coefficients in its derivative formulas is necessarily skew-symmetric.

Proposition 3.(Formula of Torsion)
The torsion $\tau(t)$ of the regular curve $\mathbf{r} = \mathbf{r}(t)$ is given by

$$\tau(t) = \frac{\Big(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t)\Big)}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|^2}.$$

Sketch of Proof: By Propostion 2, we know that

$$\gamma(t) = \frac{\mathbf{r}'(t) \times \mathbf{r}''(t)}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|},$$

so differentiate the equation above w.r.t. t and use the Frenet formula, we obtain

$$\begin{aligned}\gamma'(t) &= -\tau(t)\beta(t)\cdot\frac{\mathrm{d}s}{\mathrm{d}t} \quad \big(\text{since } \gamma'(s) = -\tau(s)\beta(s) \text{ and } s = s(t)\big)\\ &= \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathbf{r}'(t)\times\mathbf{r}''(t)}{|\mathbf{r}'(t)\times\mathbf{r}''(t)|}\right)\\ &= \frac{\mathbf{r}'(t)\times\mathbf{r}'''(t)}{|\mathbf{r}'(t)\times\mathbf{r}''(t)|} + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{|\mathbf{r}'(t)\times\mathbf{r}''(t)|}\right)\cdot(\mathbf{r}'(t)\times\mathbf{r}''(t)) \quad \big(\text{since } \mathbf{r}''(t)\times\mathbf{r}''(t) = \mathbf{0}\big).\end{aligned}$$

Using the equation that

$$\beta(t) = \frac{|\mathbf{r}'(t)|}{|\mathbf{r}'(t)\times\mathbf{r}''(t)|}\mathbf{r}''(t) - \frac{\mathbf{r}'(t)\cdot\mathbf{r}''(t)}{|\mathbf{r}'(t)||\mathbf{r}'(t)\times\mathbf{r}''(t)|}\mathbf{r}'(t)$$

and

$$\begin{aligned}-\tau(s)\cdot|\mathbf{r}'(t)| = &\left(\frac{\mathbf{r}'(t)\times\mathbf{r}'''(t)}{|\mathbf{r}'(t)\times\mathbf{r}''(t)|} + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{|\mathbf{r}'(t)\times\mathbf{r}''(t)|}\right)\cdot(\mathbf{r}'(t)\times\mathbf{r}''(t))\right)\\ &\cdot\left(\frac{|\mathbf{r}'(t)|}{|\mathbf{r}'(t)\times\mathbf{r}''(t)|}\mathbf{r}''(t) - \frac{\mathbf{r}'(t)\cdot\mathbf{r}''(t)}{|\mathbf{r}'(t)||\mathbf{r}'(t)\times\mathbf{r}''(t)|}\mathbf{r}'(t)\right)\end{aligned}$$

By some computation and use the notation that

$$\big(u, v, w\big) := u\times v\cdot w$$

we get the desired formula. ■

Corollary 1. If the space curve $C : \mathbf{r} = \mathbf{r}(s)$ is parametrized by arc-length, then the torsion $\tau(s)$ is given by

$$\tau(s) = \frac{\big(\mathbf{r}'(s), \mathbf{r}''(s), \mathbf{r}'''(s)\big)}{|\mathbf{r}''(s)|^2}.$$

Proof: Note that if the space curve $C : \mathbf{r} = \mathbf{r}(s)$ is parametrized by arc-length s, then

$$|\mathbf{r}'(s)\times\mathbf{r}''(s)| = |\alpha(s)\times\big(\kappa(s)\beta(s)\big)| = \kappa(s)|\alpha(s)\times\beta(s)| = \kappa(s) = |\mathbf{r}''(s)|,$$

which completes the proof ■

Corollary 2. The space curve $C : \mathbf{r} = \mathbf{r}(t)$ is a space curve $\iff \tau(t) \equiv 0 \iff \big(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t)\big) \equiv 0$.

Proof: By Propostion 3 and the definition of the torsion $\tau(t)$. ■